

Assignment 1

George Boufis

Task 1

Domain: Shared Housing and Roommate Matching

Scenario:

In recent years, rental prices have risen significantly compared to average salaries, making it difficult for many individuals to afford an apartment on their own. To reduce housing costs, more and more people seek to rent an apartment with a roommate, sharing both the rent and the utility bills.

However, successful co-living requires compatibility between roommates. They must respect the house rules set by the owner, communicate effectively, share household duties and be reliable in paying their share of expenses.

The scenario focuses on developing a system that helps people to find compatible roommates and suitable shared apartments based on predefined criteria such as budget, location, lifestyle, amenities of house, preferences and house rules.

A knowledge graph represents apartments, landlords, tenants, rules, amenities and compatibility relationships, budget.

A LLM allows users to ask natural-language questions. The QA-system combines both to deliver accurate, personalized and explainable answers to users searching for co-living opportunities.

Data Availability: potential data sources like Badi.com, Roomster.com etc, landlord, tenant registration databases and user-provided preferences collected through surveys or application forms.

Task 2

- What the system does

The proposed system aims to help people find apartments to rent with a compatible roommate, so that they can share the rent and all household expenses.

The system will allow users to ask natural-language questions using a GraphRAG approach, a LLM will retrieve and reason over a knowledge Graph that contains information about apartments, amenities, house rules, landlords, tenant preferences and budget.

- Who the users are:

The primary users are individuals looking for shared area accommodation - for example students, young professionals or anyone who cannot afford to rent alone.

Secondary users could be the owners of the house, who can register their apartments and define house rules and property agents who wish to promote co-living spaces.

- Why it is useful:

Rental prices are rising while salaries remain stable, making housing increasingly unaffordable.

Finding a compatible roommate is difficult because it requires evaluating not only rent and location but also personal compatibility, house rules and shared responsibilities. This QA-system addresses those issues by providing a smart, conversational assistant that filters available apartments according to the user's needs, preferences and budget, and ensures that the suggested roommate are likely to be a good match in lifestyle and reliability.

Task 3

1. Which apartments have rent below 400 euros?
2. Which apartments have Wi-Fi?
3. Which apartments have two bathrooms?
4. In which apartments does the owner forbid smoking?
5. Which apartments are located in the city center?
6. Are pets allowed to stay in the apartment?
7. Has the roommate shared an apartment before?
8. Which rules has the owner defined for the apartment?
9. Are the roommates following the rules of the owner's apartment?
10. Does the owner have other apartments for rent?
11. Are the roommates allowed to smoke in the apartment?
12. Is the apartment located near the metro station?
13. In which location is the rent the lowest?
14. How many apartments have more than one bathroom?
15. In how many apartments has the owner set up a rule that smoking is forbidden?
16. How many apartments have view to the sea?
17. How many apartments have parking?
18. How many apartments does the owner have for rent?

Task 4

Q1) The KG must include an entity type "Apartment" with an attribute "rent" that stores the monthly rental cost as a numeric value (in euros).

Q2) The KG must include an entity type "Apartment" and an entity "Amenity", linked through the relation hasAmenity(Apartment -> Amenity). Each apartment should have a list of amenities (Wi-Fi, parking, heating).

Q3) The KG must include an entity "Apartment" with an attribute "numberOfBathrooms".

Q4) The KG should represent the relationship between Owner, Apartment, Rule. The relations owns(Owner -> Apartment) and hasRule(Apartment -> Rule) are required, with an attribute or description in the Rule entity.

Q5) The KG must include an attribute "location" for each Apartment, specifying its geographic area.

Q6) The KG should include an entity "Apartment" and an attribute "allowsPets".

Q7) The KG must include entities "Roommate", "Apartment", connected through a relation such as hasShared(Roommate -> Apartment).

Q8) The KG must include the entities "Owner", "Apartment", "Rule". These entities should be connected through the relations: owns(Owner -> Apartment), definesRule(Owner -> Rule), hasRule(Apartment -> Rule).

Q9) The KG should model relationships among "Roommate", "Apartment", "Rule", "Owner". It must include owns(Owner -> Apartment), hasRule(Apartment -> Rule), followsRule(Roommate -> Rule).

Q10) The KG should include the entities "Owner", "Apartment" linked via a relation such as owns(Owner -> Apartment).

Q11) The KG must connect entities "Roommate", "Rule", "Apartment" through the relations hasRule(Apartment -> Rule), appliesTo(Rule -> Roommate).

Q12) The KG must represent a relationship between entities "Apartment" and "Metro Station", such as near(Apartment -> Metro Station).

Q13) The KG must include the entity "Apartment" with attributes rent and location.

Q14) The KG must model the entity "Apartment" with an attribute numberOfBathrooms.

Q15) The KG must include the entities "Apartment", "Owner", "Rule", relations such as owns(Owner -> Apartment), hasRule(Apartment -> Rule).

Q16) The KG must include an entity "Apartment" with an attribute viewType, hasView(Apartment -> View).

Q17) The KG must include entities "Apartment", "Amenity", connected by the relation hasAmenity(Apartment -> Amenity).

Q18) The KG must include entities "Apartment", "Owner" linked through a relation such as owns(Owner -> Apartment).

Report

Task 1:

Domain: Shared Housing

Many people today cannot afford to rent alone, so they look for shared apartments to split rent and bills. The system helps users to find compatible roommates and suitable apartments based on rules, price and amenities. Possible data sources include co-living platforms like Roomster and Badi.

Task 2:

The system allows users to ask questions in natural language such as "Which apartments have rent below 400 euros". The LLM understands the question and the Knowledge Graph provides structured data about apartment, owners, rules and amenities. It is useful for people searching for affordable and compatible co-living options.

Task 3:

Three types of questions were created:

- 1) Factual: "Which apartment have Wi-Fi?"
- 2) Relational: "Has the roommate shared an apartment before?"
"Does the owner rent other apartments?"
- 3) Analytical: "In which area is the rent lowest?"
"How many apartments have parking?"

These questions show different reasoning needs, facts, relationships and summaries.

Task 4:

The Knowledge Graph must include entities such as Apartment, Owner, Rule, Amenity, Roommate. Important relations are owns, hasAmenity, hasRule, followsRule, liveIn.

Key attributes include rent, location, numberOfBathrooms, allowsPets. The KG should support filtering, counting and comparisons so that QA system can answer the use questions correctly.